

Physics-informed neural networks: dynamics and parameter recovery for a WNT–RA–HOX model of colorectal cancer stemness

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01 Model & formulation

From signaling to stemness. Seven ODEs connect WNT, APC and retinoic-acid regulation. We study dynamics, parameter recovery and uncertainty.

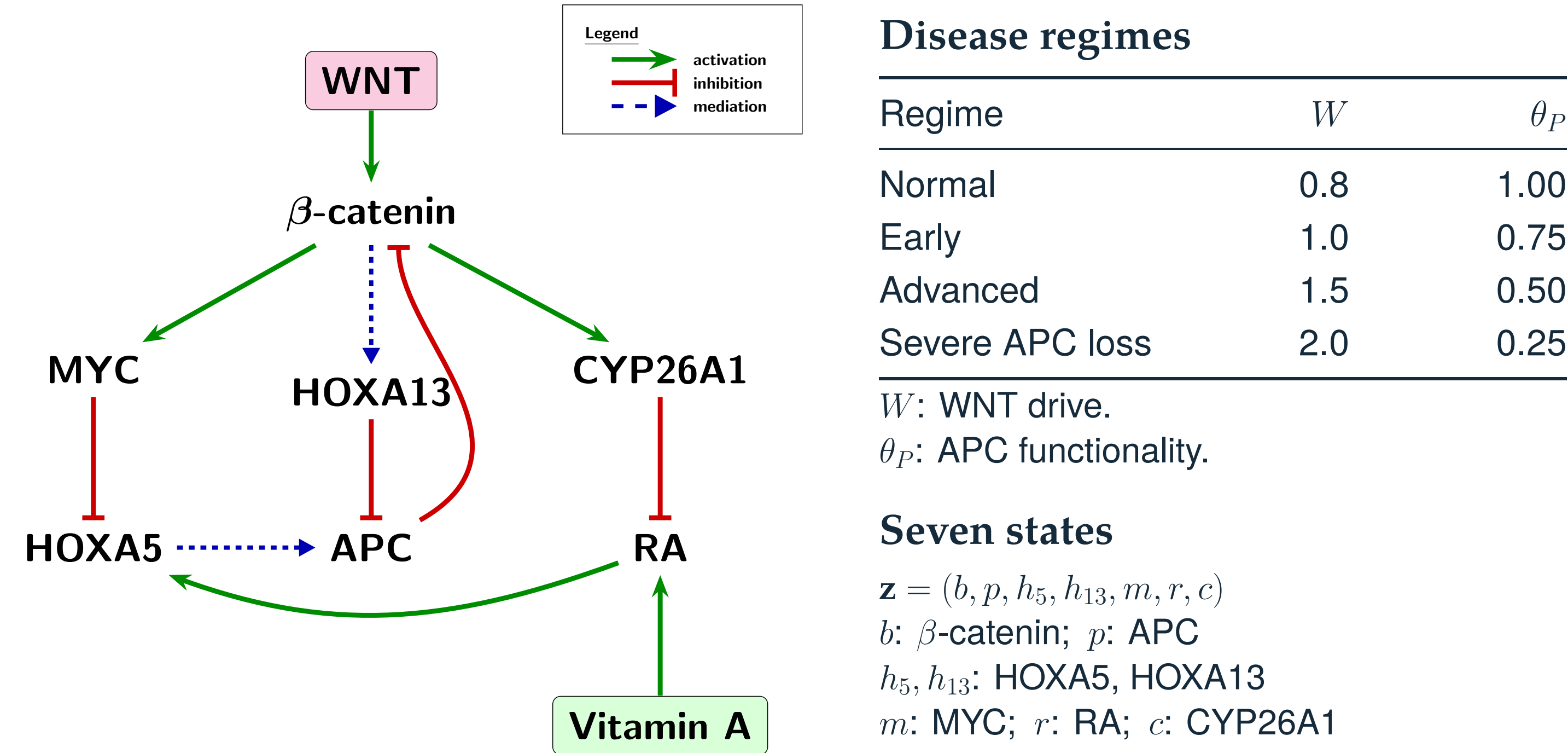


Fig. 1. Regulatory network. Green: activation; red: inhibition; blue dashed: mediation. Early and Advanced denote adenoma.

Nondimensionalization

Scale concentrations by characteristic values and time by β -catenin turnover, $d_B = 1 \text{ h}^{-1}$:

$$X = X_* x, \quad \tau = d_B t, \quad \varepsilon_X = \frac{d_B}{d_X}. \quad (1)$$

Hill activation: $H_n(u; \kappa) = u^n / (\kappa^n + u^n)$. A dot below denotes $d/d\tau$.

$$\begin{aligned} \dot{b} &= W + \eta_{13} H_{n_H}(h_{13}; \kappa_{13}) - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}, \\ \varepsilon_P \dot{p} &= \frac{1 + \rho_5 h_5}{1 + \rho_B b + \rho_{13} h_{13}} - D_P(\theta_P) p, \\ \varepsilon_5 \dot{h}_5 &= a_5 + \eta_R H_1(r; \kappa_R) - h_5 - \eta_M \frac{m h_5}{\kappa_M + m}, \\ \varepsilon_{13} \dot{h}_{13} &= a_{13} + \eta_{B13} H_{n_B}(b; \kappa_{B13}) + \eta_{M13} H_{n_M}(m; \kappa_{M13}) - h_{13}, \\ \varepsilon_M \dot{m} &= a_M + \eta_{BM} H_{n_B}(b; \kappa_{BM}) - m, \\ \varepsilon_R \dot{r} &= \mu_R(\tau) - r - \lambda_{CC} r, \\ \varepsilon_C \dot{c} &= a_C + \eta_{RC} H_1(r; \kappa_{RC}) + \eta_{BC} H_{n_B}(b; \kappa_{BC}) - c. \end{aligned} \quad (2)$$

APC loss and retinoid forcing

$$\begin{aligned} D_P(\theta_P) &= 1 + \delta_{P1}(1 - \theta_P), \\ \mu_R(\tau) &= \mu_0 + A_R \left[1 + \cos\left(\frac{2\pi\tau}{T_R} - \varphi\right) \right] \\ &\quad + \frac{D_R}{2} [\tanh q(\tau - \tau_1) - \tanh q(\tau - \tau_2)]. \end{aligned} \quad (3)$$

Reference: Radau on $\tau \in [0, 150]$; ATRA pulse $[40, 88]$. Relative timescales $\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$ produce mild stiffness.

Derived readout: stemness

$$S(\tau) = \frac{b(1 + \alpha_{13} h_{13})}{(1 + p)(1 + \alpha_5 h_5)}, \quad \alpha_5 = \alpha_{13} = 1.$$

Higher S represents greater stemness in this model.

Scope and next step

Synthetic-data study. Treatment changes only D_R ; all other forcing is retained. Fisher diagnostics are local; HMC intervals are uncalibrated. Next: reduced-parameter PINN fits and validation across noise and sampling levels.

02 Dynamics & PINN reconstruction

Normal regime: $W = 0.8$, $\theta_P = 1$; MYC and APC share each axis.

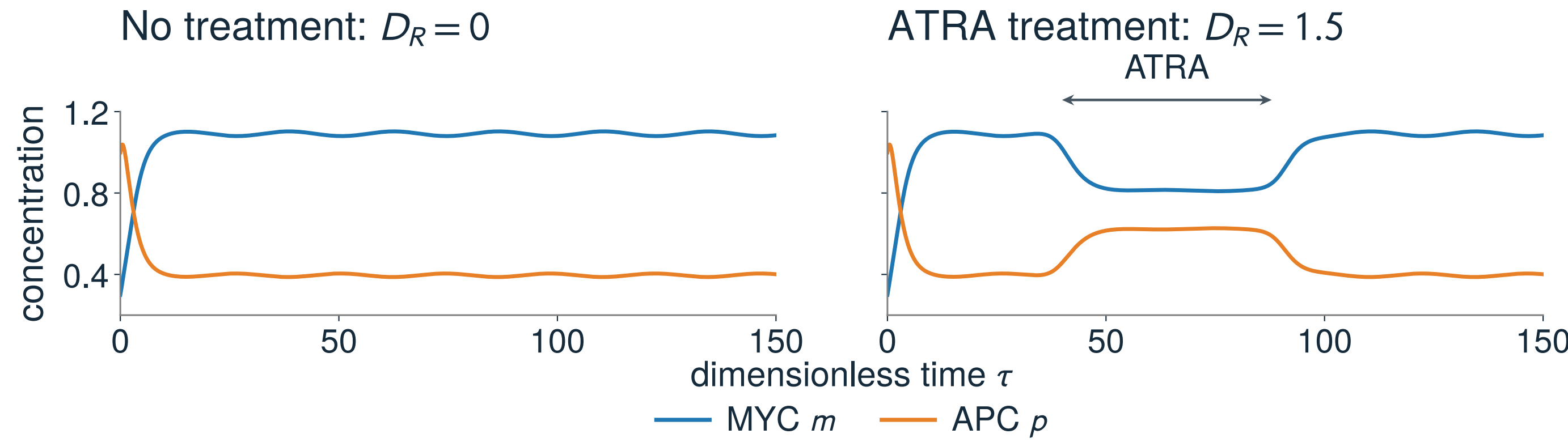
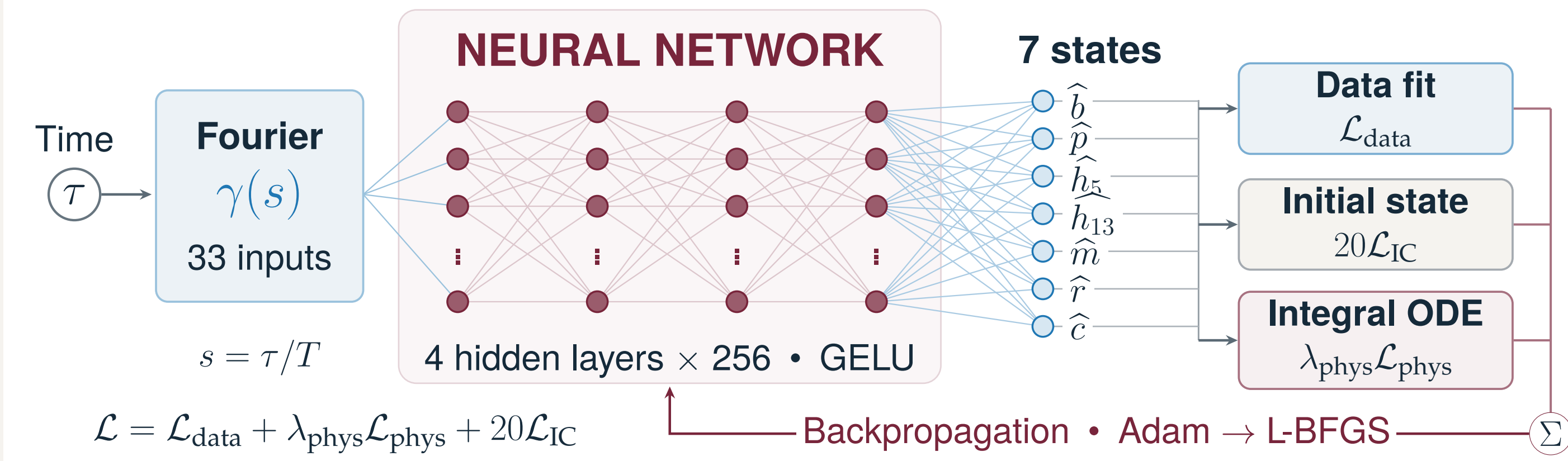


Fig. 2. Radau solutions; baseline circadian forcing $A_R = 0.04$ in both panels.



$$\mathbf{r}_k = \hat{\mathbf{z}}_{k+1} - \hat{\mathbf{z}}_k - \frac{\Delta \tau_k}{2} (\mathbf{F}_k + \mathbf{F}_{k+1}). \quad (4)$$

Ten state networks share 36 parameters; 150 times/condition ($\sigma = 0.002$). Relative state weights; adaptive λ_{phys} ; two starts.^{1,2}

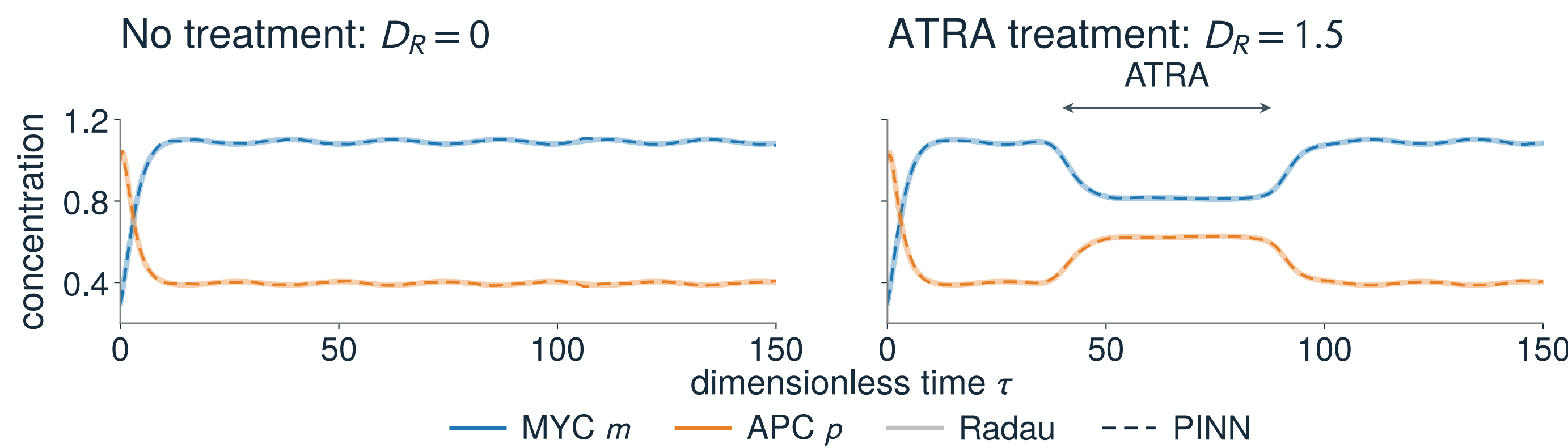


Fig. 3. Saved inverse-PINN fits (dashed) and Radau (solid). Relative L^2 errors, MYC/APC: untreated 0.31 / 0.75 %; treated 0.40 / 0.54 %.

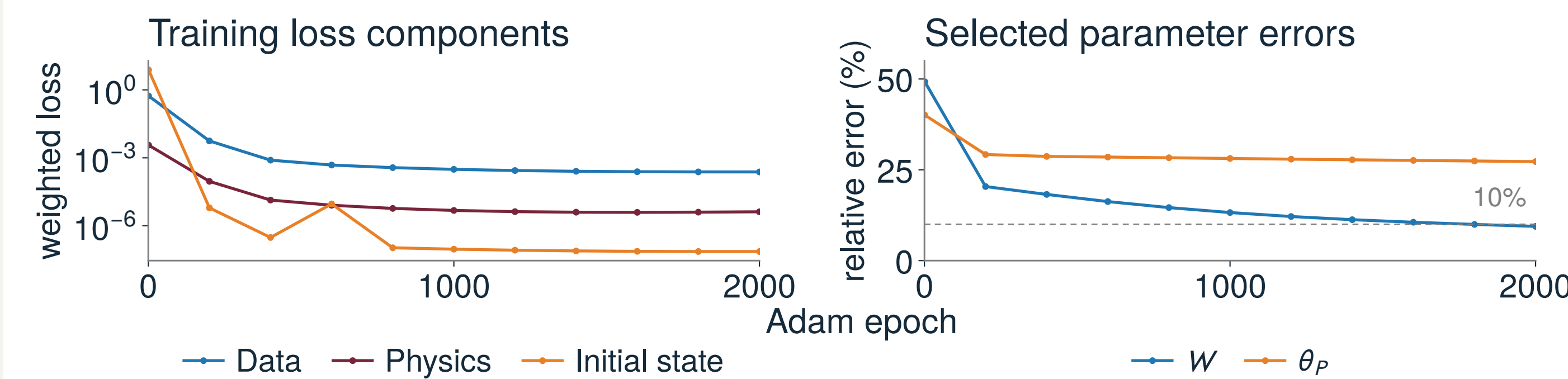


Fig. 4. Saved Adam checkpoints from the selected Normal run; losses average ten conditions. Final fits also include L-BFGS.

Parameters within 10% of truth

Method	Normal	Early	Advanced	Severe	Total
Autodiff	16	9	6	6	37
Integral	17	16	10	7	50

Table 1. Ten conditions per method: 37 $\rightarrow$ 50 of 144 parameter–regime pairs. The gain belongs to the integral implementation as a whole.

Key references

- [1] M. Raissi, P. Perdikaris and G. E. Karniadakis. Physics-informed neural networks. *J. Comput. Phys.* 378, 686–707 (2019). doi:10.1016/j.jcp.2018.10.045.
- [2] A. Yazdani et al. Systems biology informed deep learning. *PLoS Comput. Biol.* 16, e1007575 (2020). doi:10.1371/journal.pcbi.1007575.
- [3] L. Yang, X. Meng and G. E. Karniadakis. B-PINNs. *J. Comput. Phys.* 425, 109913 (2021). doi:10.1016/j.jcp.2020.109913.

03 Identifiability & Bayesian uncertainty

Fisher information and parameter recovery

Local sensitivities at truth: 7 states, 10 conditions and 200 times.

$$\mathcal{I} = \sigma^{-2} J^T J, \quad J_{ki} = \frac{\partial y_k}{\partial \log \theta_i}, \quad \sigma = 0.002. \quad (5)$$

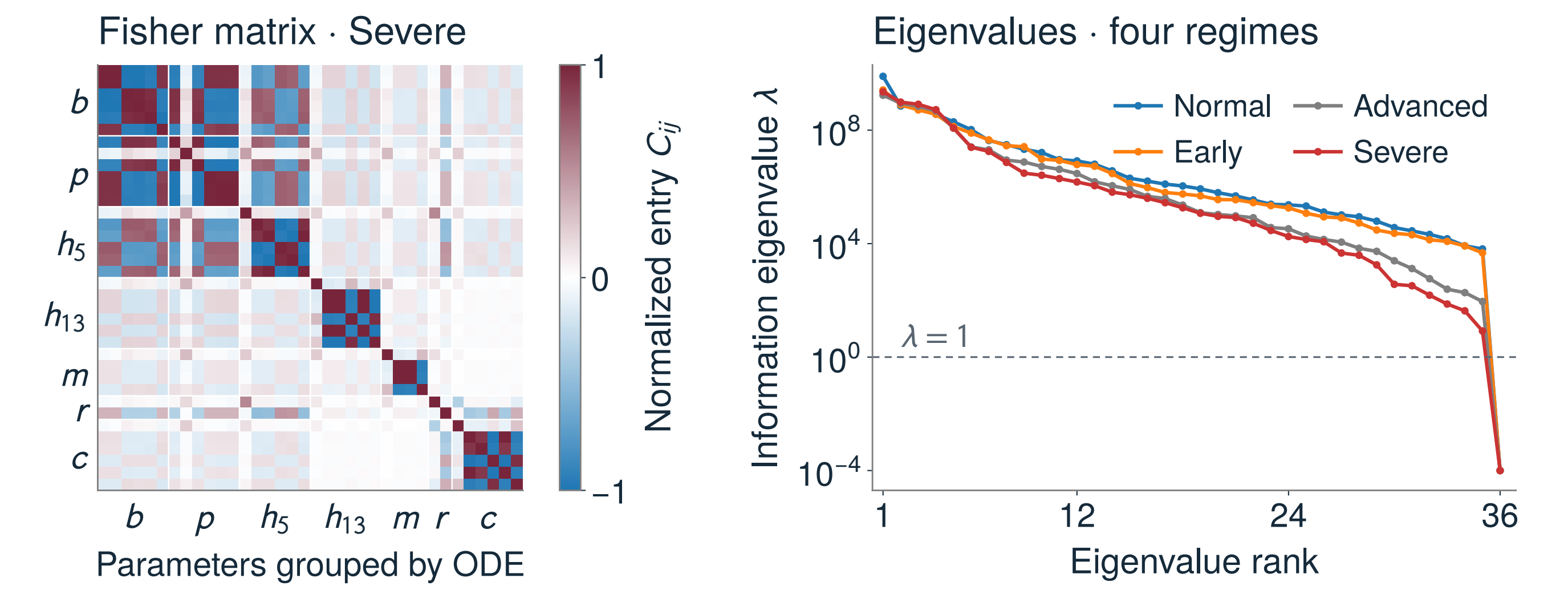
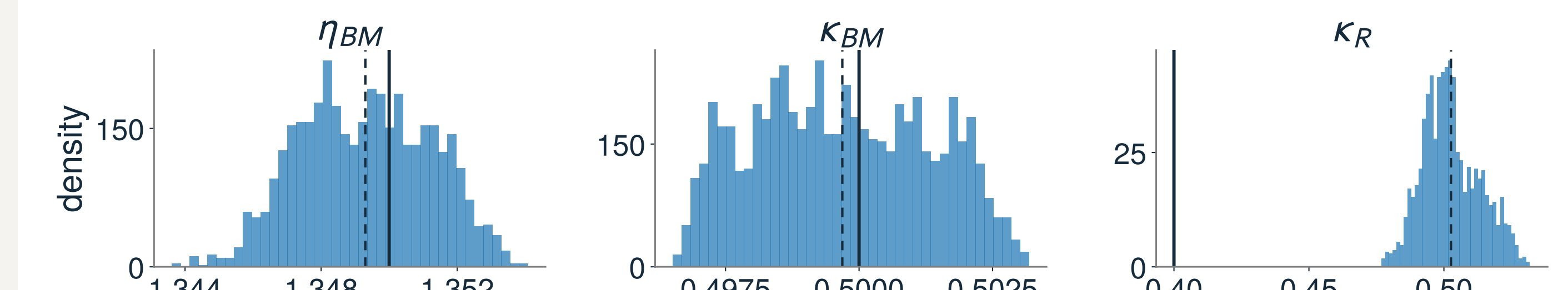


Fig. 5. Severe-regime $C_{ij} = \mathcal{I}_{ij} / \sqrt{\mathcal{I}_{ii} \mathcal{I}_{jj}}$: $|C_{ij}| \approx 1$ signals overlapping sensitivities. Each regime has **one near-null direction** ($\lambda < 1$); plotting floor 10^{-4} .

Bayesian parameter distributions

Frozen-state HMC samples all 36 parameters using the integral likelihood.³

Normal



Severe APC loss

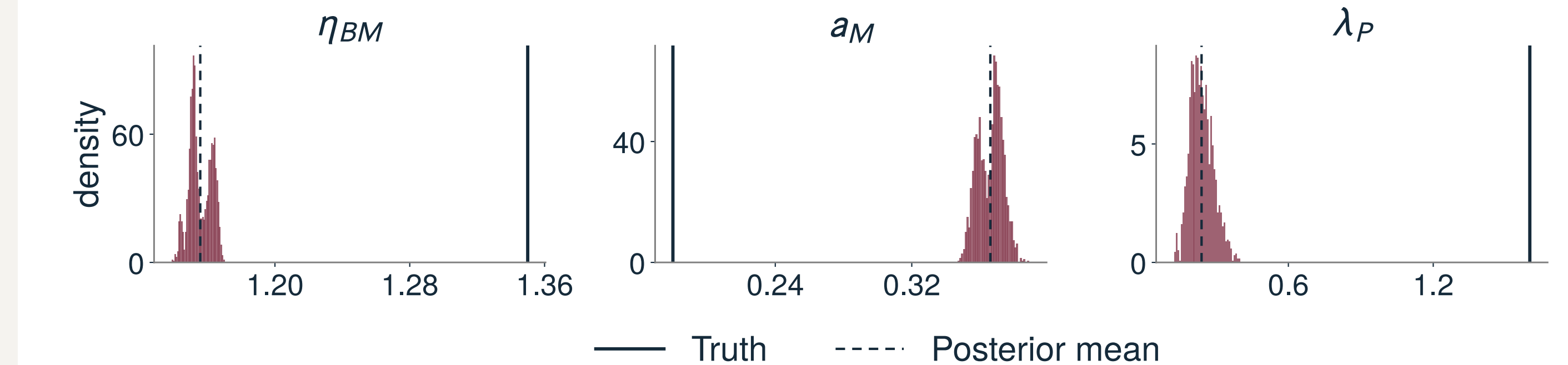


Fig. 6. Presentation marginals (2000 draws/regime): narrow distributions can miss truth.

Calibration limit. HMC fails effective-sample-size and coverage checks; interval widths are unreliable.

Fisher analysis of eight parameters

We varied eight parameters and fixed the other 28 at their true values:

$$\lambda_C, \eta_{BM}, \kappa_{BM}, \kappa_{BC}, \eta_{BC}, \eta_{RC}, \kappa_{B13}, \kappa_R$$

Fisher result	Normal	Early	Advanced	Severe
Locally identifiable	8/8	8/8	8/8	8/8
Eigenvalues below 1	0	0	0	0
cond($\mathcal{I}$)	99.7	120.9	229.4	665.2

Table 2. The observations match the full Fisher analysis above. PINN recovery of these eight parameters has not been tested.